University of Barishal

Department of CSE

3rd Year 2nd semester mid-term Exam-2024

Course name: Mathematical Analysis for Computer Science; Course code: CSE-3201

Total Marks: 25 Time: 1 hr

Answer all the questions. Marks are indicated at the end of the question

Question 5 will be considered as class test

Q1. Define random variable with example. What are the different types of random variables? Explain with appropriate example. [5]

Q2. Define index set, state space and process with independent increment. [5]

Q3. Consider the process $X(t) = A_1 + A_2 t$ where A_1 *and* A_2 are independent random variables with $E(A_i) = a_i, V(A_i) = \sigma_i^2$; $i = 1, 2$. Test the stationarity. [5]

Q4. What is martingale process? Let $\{Z_i\}$; $i = 1, 2, \cdots$ be a sequence of *i.i.d.* random variables with $E\{Z_i\} = 1$ and let $X_n = \prod_{i=1}^{n} Z_i$, then show that $\{X_n\}_{n=1}^{\infty}$ is a martingale. [5]

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Q5. Briefly describe each section of stochastic process with practical life uses that you know. [5]

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